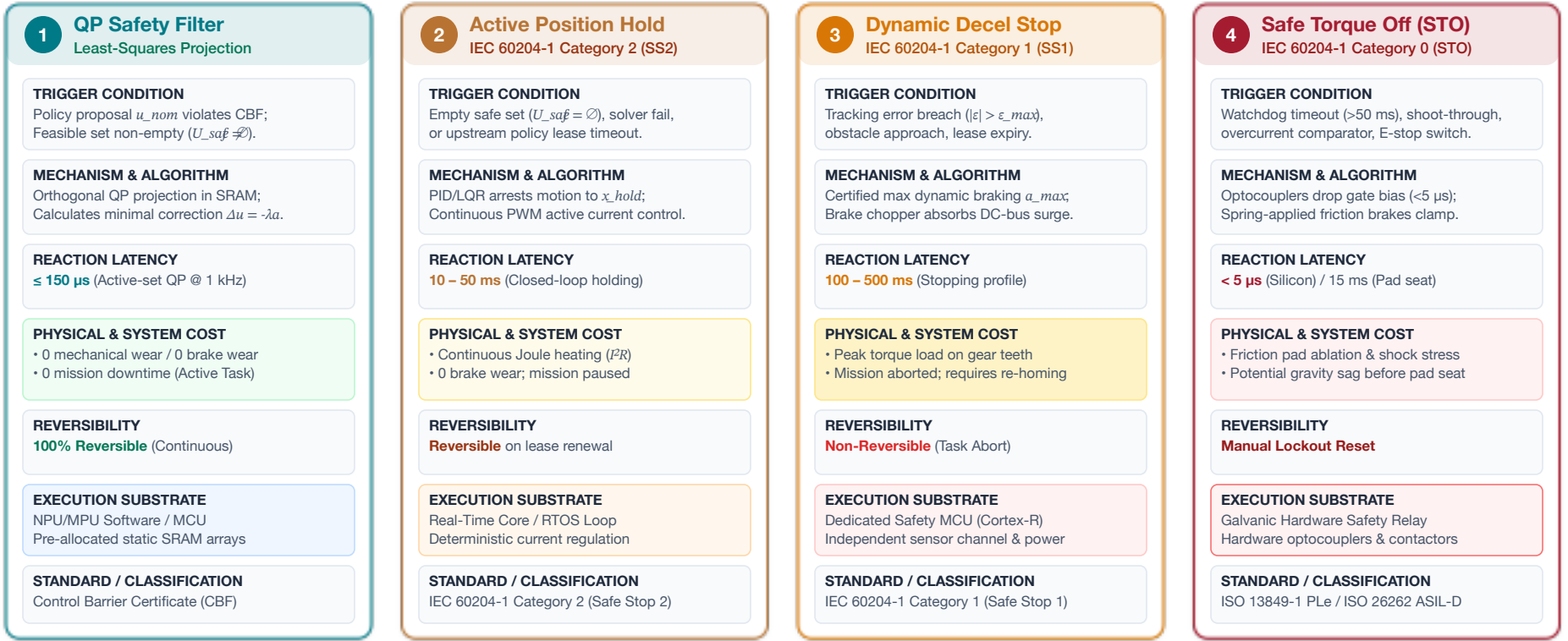


THE DETERMINISTIC FOUR-TIER FALLBACK ESCALATION LADDER

Graded Safety Authority: Escalating Physical Cost, Decreasing Software Dependency & Increasing Isolation



← Low Mechanical Cost / High Reversibility

ESCALATION OF DEFENSIVE INTERVENTION

High Hardware Cost / Non-Reversible →

SUBSTRATE INDEPENDENCE: Shared MPU / RTOS Core → Isolated Safety Co-Processor → Dual-Channel Safety MCU → Galvanic Analog Hardware / E-Stop

Systems Architectural Rule: Lower rungs must execute on substrates physically isolated from upstream software failure modes.